ME 3300 Lab-07: Strain Gauge and Error Propagation

Dr. Christopher Bitikofer

Contents

1 Learning objective **2**

2 Required Equipment **2**

3 Introduction **2**

4 Objectives **2**

5 Laboratory instruction **3**

5.1 Part 1: Build and Test a Half Arm Wheatstone Bridge Circuit . . . 3

5.2 Part 2: Preparing for analysis of the Strain and Stress Data. . . . 5

5.3 Part 3: Take Load and Strain Data for the Cantilevered Beam . . . 5

5.4 Part 4: Perform linear regression of stress vs strain to estimate Young's modulus of the beam. . . . 6

5.5 Part 5: Collect uncertainties for each variable used to determine Young's Modulus. . . . 7

5.6 Part 6: Compare uncertainty analysis from two approaches . . . 8

6 Post-Laboratory **8**

1 Learning objective

The objectives of this experiment are to provide the student with an opportunity to:

1. Learn about using strain gauges with a Wheatstone bridge
2. Learn about the uncertainty propagation and analysis
3. Learn about using an instrumentation amplifier

2 Required Equipment

1. Cantilevered beam with mounted strain gauges.
2. AD622 Instrumentation Amp
3. Powered Breadboard & ± 15 Volt source
4. Table-top Digital Multi-Meter
5. Wire & Resistor Kits
6. Screw Driver, Pliers, Wire Stripper
7. Precision Matched Resistors (350 Ω)
8. Calipers
9. Measuring Tape
10. Weight Set

3 Introduction

Measurement of strain is a fundamental tool used by engineers in both research and design. Strain measurements are often used in systems where force or pressure measurements are required. In some applications the strain (and associated stress) of a part can only be determined by measurement, as calculation is either too complicated or not accurate enough. In this experiment, you will learn to use a metallic strain gauge to measure strain in an end-loaded cantilevered beam. Simultaneously, you will calculate the stress due to bending caused by the end load. You will use these values to calculate a stress value. Finally, your stress and strain plot will be used to determine the modulus of elasticity (or Young's modulus) of the beam.

4 Objectives

- Assemble a **Wheatstone half bridge** circuit to measure a signal from two strain gauges affixed to a cantilevered beam.
- Collect enough strain and loading data to perform a linear regression to estimate Young's modulus, E, of the cantilevered beam. Compare the experimental value with published values.

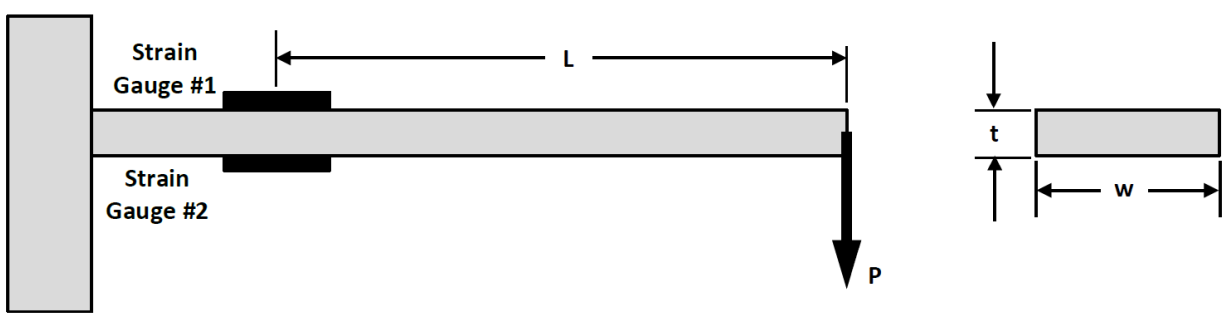

Figure 1: End loaded cantilever beam with two bonded strain gauges.

Parts 1-4 first week, Parts 5-6 second week

5 Laboratory instruction

5.1 Part 1: Build and Test a Half Arm Wheatstone Bridge Circuit

1. On the breadboard:
   - Build a half arm Wheatstone bridge circuit as shown in Figure 2.
   - Use the two strain gauges mounted to the cantilevered beam as resistors R#1 and R#2.
   - Use matching precision resistors (350Ω) for R#3 and R#4.
2. Use a 102 Ω resistor to set AD622 amplifier gain to G ≈ 500.
   - The outputs from the bridge are the inputs to the instrumentation amplifier.
   - A gain of 500 will amplify the voltage to a range suitable for the multi-meter and data acquisition.
   - Connect the output of the instrumentation amplifier to the NI-MyDAQ.
3. Connect the excitation ground to the REF pin on the instrumentation amplifier (2).
4. Provide the amplifier and the bridge with a 5V excitation voltage from Ni MyDAQ (2).
5. Use the DMM to measure the amplified output voltage, $V_o \times G$, from the instrumentation amplifier to check that your experimental setup is working properly.

Q: What extraneous effect does this Wheatstone bridge configuration compensate for?

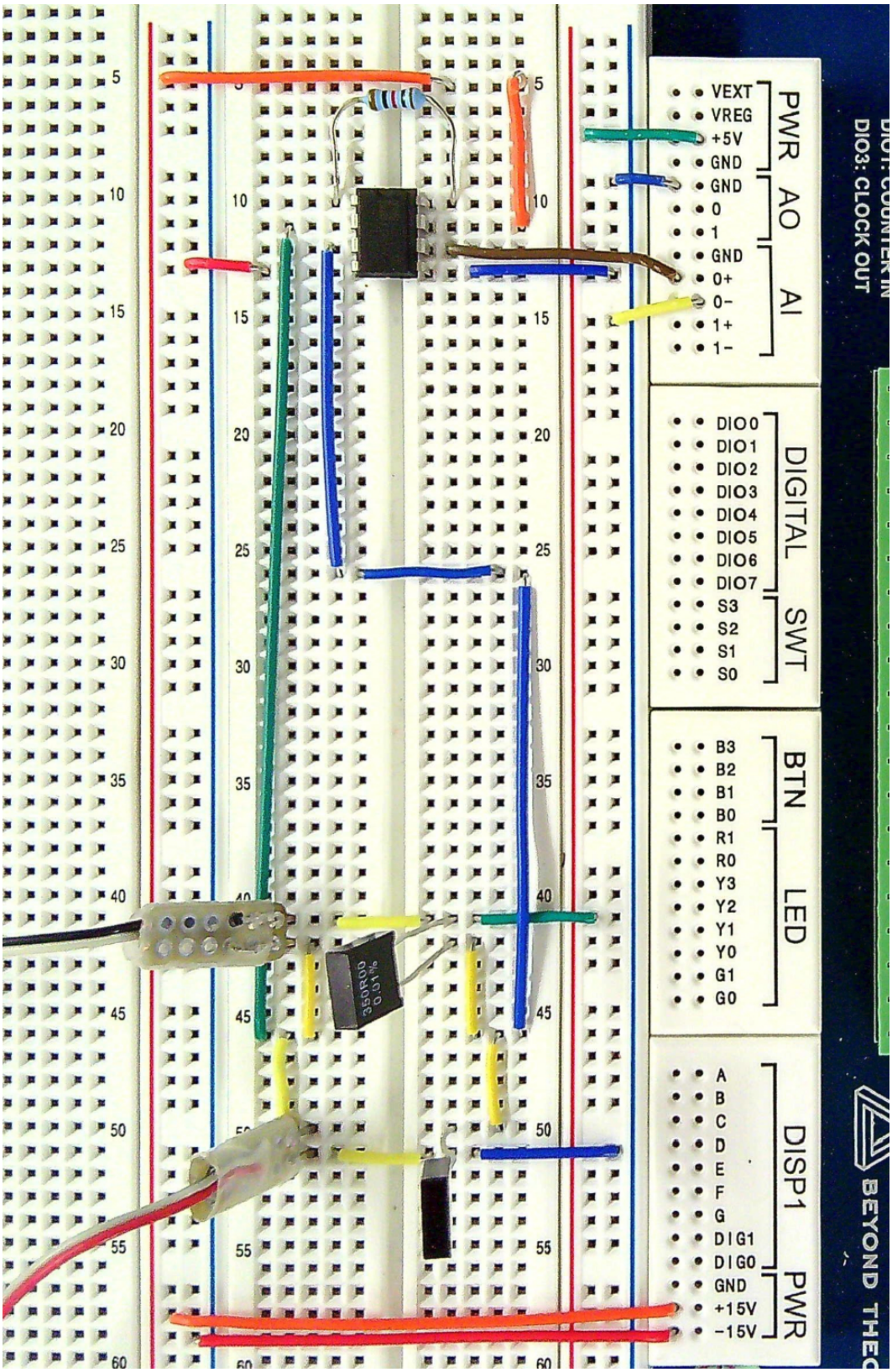

Figure 2: Two strain gauges mounted on a cantilevered beam and a half-arm Wheatstone bridge circuit.

5.2 Part 2: Preparing for analysis of the Strain and Stress Data.

1. Measure the beam using the following tools/dimensions:
   - Use a tape measure to measure the distance between the center of the strain gauge and the point of loading, L. Note the resolution of the tape measure in meters.
   - Use a vernier caliper to measure the width of the beam, w. Note the resolution of the vernier caliper in meters.
   - Use a micrometer to measure the beam thickness, t. Note the resolution of the micrometer in meters.
2. Determine local gravity using this local gravity website (use the Latitude value of 46.7324 and the elevation of 2579 ft).
3. From solid mechanics we get the relationship $\sigma = My/I$. You will use this relationship for stress calculations. The stress equation, including all the measured variables, is provided in equation 4.

$$\begin{aligned} \sigma &= \frac{My}{I}, && (1) \\ &= \frac{PL(t/2)}{wt^3/12}, && (2) \\ &= \frac{6PL}{wt^3}, \text{and} && (3) \\ &= \frac{6mgL}{wt^2}. && (4) \end{aligned}$$

4. The measured strain, ε, may be found from the half-bridge Wheatstone circuit

$$\begin{aligned} \frac{\delta V_o}{V_i G} &= \frac{GF\varepsilon}{2}, && (5) \\ \varepsilon &= 2\frac{\delta V_o}{V_i G} \times \frac{1}{GF} && (6) \end{aligned}$$

   where V_i is excitation voltage, $GF = 2.012$ is the gauge factor, G is the instrumentation amplifier gain, and δV_o is the change in output voltage from loaded to unloaded condition.

Q: What is hysteresis, and how can you mitigate it while running experiments?

5.3 Part 3: Take Load and Strain Data for the Cantilevered Beam

1. Update the MATLAB code, **MATLAB_Data_Acquisition.m**, with the instructions provided in this section.
2. Collect the single-channel voltage output from ("AI0").
   - Duration: 10 seconds
   - Sample rate: 50 samples per second
3. Set up data acquisition for single channels of DAQ system.
   - "AI0": output from the instrumentation amplifier

   a. Running your program:
      - Save a data file using this naming convention: **Mass##_@@.csv**
      - Change ## to the current mass in kilogram.
      - Change @@ to either loaded or unloaded
        - Loaded scenarios have the weight included during the measurements.
        - Unloaded at measured before and after loaded scenarios when the weight is removed.
4. Record the average voltage value in the table provided (Table 1).
5. Method of measuring strain data for each weight, example for 0.4 kg:
   (a) Record voltage measurement for unloaded scenario($[V_{i,1}]_{avg}$.)
   (b) Load 0.4 kg weight, record voltage measurement for loaded scenario ($[V_{i,2}]_{avg}$.)

(c) Record a second voltage measurement for unloaded scenario ($[V_{i,3}]_{avg}$.)

- Your 2nd column should have two unloaded voltage values.

(d) Subtract the unloaded from loaded: $G\delta V_{o,1} = [V_{i,2}]_{avg} - [V_{i,1}]_{avg}$ and $G\delta V_{o,2} = [V_{i,2}]_{avg} - [V_{i,3}]_{avg}$.

- Your 3rd column should have two $G\delta V_o$ values.

(e) Average the two values: $[G\delta V_o]_{avg} = \frac{G\delta V_{o,1}+G\delta V_{o,2}}{2}$

6. Repeat the process for the remaining weights.
7. NOTE: Make sure **not to touch** the wires in your setup since the gain of 500 will make it sensitive to any changes.

Table 1: Table to record your experimental data.

	Mass (kg)	Loaded voltage (v)	Unloaded voltage (v)	$G\delta V_o$	$[G\delta V_o]_{avg}$	ε [Use Eq.6]	σ (GPa) [Use Eq.4]
1	0.0						
2	0.4						
	0.4						
3	0.6						
	0.6						
4	0.8						
	0.8						
5	1.0						
	1.0						
6	1.2						
	1.2						
7	1.4						
	1.4						
8	1.6						
	1.6						
9	1.8						
	1.8						
10	2.0						
	2.0						

5.4 Part 4: Perform linear regression of stress vs strain to estimate Young's modulus of the beam.

1. Use the provided MATLAB code (**StrainStressPlot.m**) to obtain the for curve fitting.
2. The code calculates the stress and strain for all data points and can be used to fill in your table.
3. Using polyfit in MATLAB to perform a linear regression. The slope of the stress vs. strain plot is Young's modulus.

Q: What is the range of Young's modulus for published steel values? Does your result make sense compared to published values?

Parts 5-6 (Second Week)

5.5 Part 5: Collect uncertainties for each variable used to determine Young's Modulus.

1. Determine the uncertainty for each variable used to calculate stress and strain in the experiment. Complete the 1st two rows in Table 2. For those measurements performed in the lab (e.g., L , w , t), estimate the uncertainty. For the measurement equipment, refer to the manufacturer's online datasheets.
2. Perform this analysis for three different masses m = 0.5 kg, 1.0 kg, and 1.5 kg.
3. Use 1% uncertainty in the mass
4. Use 0.001% uncertainty in acceleration due to gravity
5. Use 1% uncertainty in the GF
6. For uncertainty in the op-amp gain (G): use the propagation of error approach, with only the source of uncertainty being from the resistance used to set the gain on the instrumentation amplifier.
7. Uncertainty in the excitation voltage (V_i) is the resolution of the measurement system used
8. Uncertainty in $G\delta V_o$: Uncertainty from resolution of the data acquisition system and zero-order uncertainty.

Table 2: Table to calculate uncertainty in Young's modulus using error propagation approach. Fill in the separate tables for m = 0.4, 1.0, and 1.4 kg.

	m (kg)	g (m/s^2)	L(m)	t(m)	w(m)	GF	G	V_i (v)	$[G\delta V_o]_{avg}$ (v)
Variable mean value	0.4								
Variable uncertainty									
Uncertainty propagated to Young's modulus (GPa)									
Total uncertainty in Young's modulus (GPa)									

	m (kg)	g (m/s^2)	L(m)	t(m)	w(m)	GF	G	V_i (v)	$[G\delta V_o]_{avg}$ (v)
Variable mean value	1.0								
Vairable uncertainty									
Uncertainty propagated to Young's modulus (GPa)									
Total uncertainty in Young's modulus (GPa)									

	m (kg)	g (m/s^2)	L(m)	t(m)	w(m)	GF	G	V_i (v)	$[G\delta V_o]_{avg}$ (v)
Variable mean value	1.4								
Vairable uncertainty									
Uncertainty propagated to Young's modulus (GPa)									
Total uncertainty in Young's modulus (GPa)									

5.6 Part 6: Compare uncertainty analysis from two approaches

- To estimate uncertainty in Young's Modulus, you will use two approaches and compare results
  1. Propagation of error Use the equation 7.
  $$E = \frac{3mgLV_iG(GF)}{wt^2\delta V_o}. \tag{7}$$
  2. Error from curve fit. The standard error of the slope can be found according to:
  $$s_{a1} = s_{yx}\sqrt{\frac{1}{\Sigma_{i=1}^{N}(x_i - \overline{x})^2}}, \tag{8}$$
  where s_{yx} is the standard error of the fit.

Q: In error propagation, which variables/uncertainties caused the most uncertainty in Young's modulus? Knowing this, how could the certainty of your results be improved?

Q: How do your error propagation and curve fit error results compare to each other? What might explain any differences?

6 Post-Laboratory

For this experiment, submit the following items on Canvas for your post-laboratory assignments.

Week 1:

1. Stress vs. Strain plot with linear regression.
   - Submit a single pdf figure with stress (in MPa) plotted vs strain (in micro-strain).
     - Using the "scatter" plot function, raw data points should be with `MarkerSize = 90`.
   - Include the linear regression line, of which the slope is Young's modulus.
   - Use the text command to place your experimentally determined value (in GPa) for Young's modulus on the plot.
   - The plot should be 6" tall and 6.5" wide.
   - Properly annotate your plot: axis labels, title, legend, etc.
     - Use title: "FirstName LastName's Stress vs Strain".
   - Set axis grid lines on, box off, and figure background color to white.
2. A straindata_CB.m file. Save the files with your initials on the end and submit it on Canvas. Use your initials, not mine!
3. Answer all questions in the post-lab assessment on Canvas.

Week 2:

1. Submit the pdf file with your table for all hand calculation uncertainty propagation.
2. Answer all questions in the post-lab assessment on Canvas.